11 Synthesis problems, reasoning and starter routines

Sheet 62 Tue 27 Apr. 2027 *Synthesis problems across several areas of the curriculum*

1. Warm-up · Quick [EQ] Solve $5x - 3 = 2x + 9$.

   Ans. $3x = 12$, $x = 4$.

2. Core · Problem [PRO] A price rises by 20%, then a fixed discount of 1,000 Ar is applied, giving 11,000 Ar. What was the original price?

   Ans. $1.2p - 1{,}000 = 11{,}000$, $p = \dfrac{12{,}000}{1.2} = 10{,}000$ Ar.

3. Core + · Method [REG] $f(x) = 2x + 1$ and $g(x) = -x + 7$. Find the point of intersection of the curves, then solve $f(x) < g(x)$.

   Ans. $2x + 1 = -x + 7$: $x = 2$, point $(2, 5)$; $f(x) < g(x) \Leftrightarrow 3x < 6 \Leftrightarrow x < 2$.

4. Review · Quick [QP] A recipe for 4 people needs 250 g of flour. How much for 6 people?

   Ans. $250 \div 4 \times 6 = 375$ g.

5. Thread · Quick [GEO] A shape of area 4 cm^2 is enlarged with scale factor 3. What is the area of the enlargement?

   Ans. Area $\times 3^2$: 36 cm^2.

   Pitfall: multiplying the area by 3 instead of 3^2.

Sheet 63 Thu 29 Apr. 2027 *Proof by contradiction and by cases*

1. Warm-up · Quick [DIV] What is the parity of $n(n+1)$ for an integer n?

   Ans. Always even: one of the two consecutive integers is even.

2. Core · Proof by cases [DEM] Show that $n(n+1)$ is even for every integer n.

   Ans. If n is even, the product contains an even factor. If n is odd, $n + 1$ is even. In both cases, the product is even.

3. Core + · Contradiction [DEM] Show by contradiction that the sum of a rational number r and an irrational number a is irrational.

   Ans. Suppose $r + a = q \in \mathbb{Q}$; then $a = q - r \in \mathbb{Q}$ (difference of two rationals): contradiction with a being irrational.

4. Review · Formula [IMG] $g(x) = 2x + 5$. Image of -1? Preimage of 11?

   Ans. $g(-1) = 3$; $2x + 5 = 11 \Leftrightarrow x = 3$.

5. Thread · Formula → graph [REG] Plot the points with abscissae -2 ; -1 ; 0 ; 1 ; 2 on the curve of $f(x) = x^2$; give their ordinates.

   Ans. 4 ; 1 ; 0 ; 1 ; 4: the points are symmetric about the vertical axis.

Sheet 64 Mon 3 May 2027 *Logical connectives, contrapositive and counter-example in context*

1. Warm-up · Quick [DEM] Write the negation of "$x > 2$" and then of "$x > 2$ or $x < -1$".

   Ans. $x \leqslant 2$; $x \leqslant 2$ and $x \geqslant -1$, that is $-1 \leqslant x \leqslant 2$.

2. Core · Method [DEM] Write the contrapositive of "if n^2 is even, then n is even".

   Ans. "If n is odd, then n^2 is odd" (equivalent to the original statement).

3. Core + · Counter-example [DEM] Disprove: "every prime number is odd".

   Ans. Counter-example: 2 is prime and even.

4. Review · True/false [PNP] True or false? Justify. The perimeter of a square is proportional to the length of its side.

   Ans. True. $P = 4c$: constant of proportionality 4.

5. Thread · Reading code [ALG] What does $h(3)$ return? Write $h(x)$ as an expression.

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def h(x):
    y = x + 2
    return y * y
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Ans. $h(3) = 5 \times 5 = 25$; $h(x) = (x+2)^2$.